Training Assignment 07

NUMA01: Computational Programming with Python

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This assignment has 5 tasks.
This tasks trains various aspects of arrays in connection with some methods of numerical linear algebra. It is based on Chapter 4 and 5 of the course book.

Warming-up Exercises

Task 1

Read Chapter 5 of the course book.

Task 2

A student has produced the following code for implementing a matrix-vector product. Help the student to make the code run properly. After you provided a improved version the student wants to know why your code is correct. With confidence you explain why your version is correct!! Hint: Each line of `matvec` contains one or more mistakes.

```
from numpy import *

# returns y = A * x
def matvec( A, x ):
    # make sure A has as many columns as x has entries
    y = x
    assert A.shape[1] == len(x)
    for r in range(1,A.shape[1]):
        for c in range(1,A.shape[0]):
            y[r] = A[c, r] * x[r]
    return y

x = array([1., 2., 3])
M = array([[1., 2, 3.],
           [4., 5., 6]])

z = matvec( M, x )
# is this correct?
print(z)
```

Exercises

Task 3

A central theorem in linear algebra says that the eigenvalues of real symmetric matrices are real and that their eigenvectors are orthogonal. Try to verify this theorem by using square random matrices of dimensions not less then 5.
Test your function.

Task 4

For $n \in \mathbb{N}$ construct a $n \times n$ matrix A with the properties

$$\begin{aligned} A_{i,i} &= -2, \quad i = 1, \ldots, n \\ A_{i+1,i} &= 1 \quad i = 1, \ldots, n-1 \\ A_{i,i+1} &= 1 \quad i = 1, \ldots, n-1 \end{aligned}$$

and all other elements being zero.
Do this task by using the `numpy` command `diag` or `eye` or alternatively, by using for-loops. Write a function, that takes an integer n and returns such a matrix.

Task 5

Create a vector `x=linspace(0,2*pi,500)` and a vector `u = sin(x)`. Create the Matrix D by calling the function from the previous task with n being the length of vector u. Compute $y = D\,u$ and plot the result. Compare with a plot of the second derivative of $sin(x)$.